Nearby stars in J-PLUS DR4 × Gaia DR3

Colour–colour diagrams of 1,089,852 stars within 500 pc with parallax errors below 10% · 5 November 2026

Summary

Every J-PLUS DR4 source with a Gaia DR3 counterpart was selected where the parallax implies a distance below 500 pc ($\varpi > 2$ mas) with a relative parallax error below 10%, so the distance can be obtained by simply inverting the parallax. The J-PLUS quality cuts FLAGS < 4, MASK_FLAGS = 0 and NORM_WMAP_VAL > 0.8 were applied in every filter used, with the aperture-corrected 3″ magnitudes (MAG_APER_COR_3_0). The selection runs entirely on CEFCA's TAP service in ADQL, because J-PLUS provides a pre-computed Gaia DR3 cross-match table that already carries the parallaxes.

Quantity	Value
J-PLUS tiles covered	2,563 (all of DR4)
Gaia-matched rows with $\varpi > 2$ mas and $\sigma_{\varpi}/\varpi < 0.1$	1,553,253
…passing the J-PLUS cuts in g, r, i, J0660	1,089,884
…after keeping one row per Gaia source	**1,089,852**
Median distance	346 pc
Stars within 100 pc	16,966
r-band range (1st–99th percentile)	13.0 – 20.7 mag
Fraction with Gaia RUWE < 1.4	93%

1 Data and access

- **Service:** CEFCA TAP, `https://archive.cefca.es/catalogues/vo/tap/jplus-dr4`. DR4 was used because it is the most recent release with a TAP endpoint; DR3 is also available (`…/jplus-dr3`) and works with the same query.
- **Photometry:** `jplus.MagABDualObj`: dual-mode catalogue (detection in r, photometry in all 12 filters). The photometric columns are 12-element arrays, one element per filter.
- **Astrometry:** `jplus.xmatch_gaia_dr3`: cross-match on (tile_id, number), already containing `plx`, `e_plx`, `ruwe`, proper motions and Gaia photometry, so no join with a separate Gaia table is needed.

2 Selection

Cut	Value	Purpose
Gaia parallax	plx > 2 mas	d = 1000/ϖ < 500 pc
Parallax precision	e_plx < 0.1 × plx	distance from 1/ϖ with negligible bias
SExtractor flags	FLAGS < 4	no saturation / truncation; blending allowed (flags 1, 2)
J-PLUS masks	MASK_FLAGS = 0	outside masked regions (bright stars, artefacts)
Weight map	NORM_WMAP_VAL > 0.8	well-exposed region of the tile
Magnitudes	MAG_APER_COR_3_0	3″ aperture, corrected to total

The parallax cuts are applied in ADQL. The J-PLUS cuts are applied per filter, in every filter that enters the plots (gSDSS, rSDSS, iSDSS, J0660), after download. Stars that appear in two overlapping tiles were reduced to one entry per Gaia source, keeping the closest match.

3 ADQL query

```
SELECT m.tile_id, m.number, m.alpha_j2000, m.delta_j2000,
       m.flags, m.mask_flags, m.norm_wmap_val,
       m.mag_aper_cor_3_0, m.mag_err_aper_cor_3_0,
       x.source, x.angdist, x.plx, x.e_plx, x.ruwe, x.gmag, x.bp_rp
FROM jplus.MagABDualObj AS m
JOIN jplus.xmatch_gaia_dr3 AS x
  ON x.tile_id = m.tile_id AND x.number = m.number
WHERE x.plx > 2 AND x.e_plx < 0.1 * x.plx
  AND m.tile_id >= <lo> AND m.tile_id < <hi>      -- one of 8 chunks
```

Run as an asynchronous TAP job (pyvo). The last line splits the sky into 8 tile_id ranges; see §4.

4 Practical notes (things that would have gone wrong)

- **Column names.** The corrected aperture magnitude is `mag_aper_cor_3_0` (not `mag_aper_3_0_cor`), and Gaia parallaxes are `plx` / `e_plx` in the cross-match table.
- **Filter order in the arrays is not wavelength order.** Element k of every vector column corresponds to `jplus.Filter.filter_id` = k+1: rSDSS, gSDSS, iSDSS, zSDSS, uJAVA, J0378, J0395, J0410, J0430, J0515, J0660, J0861. Assuming wavelength order silently produces wrong colours.
- **1,000,000-row limit.** The service returns at most 10^6 rows per job, without an error. The first full-sky query returned exactly 10^6 rows covering only 1,707 of 2,563 tiles. Splitting by tile_id into 8 chunks (≈ 180–230 k rows each) recovers the full sample, and the chunks ran faster (2–4 min each, 4 in parallel) than the single query (≈ 15 min).

5 Results

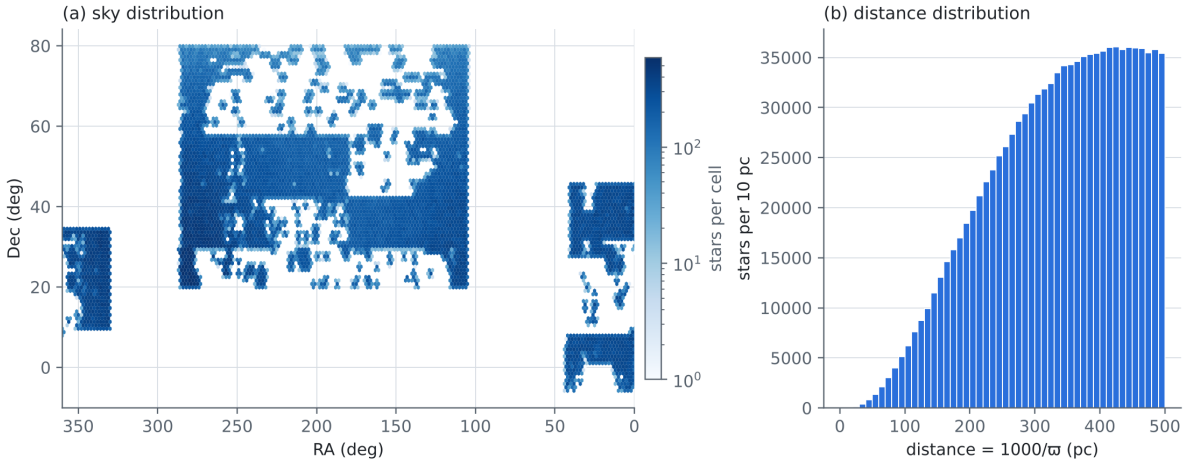


Figure 1. (a) Sky distribution of the final sample, which follows the J-PLUS DR4 footprint. (b) Distance distribution. The counts grow roughly as d² out to ~300 pc and then flatten: beyond that the 10% parallax-error cut and the J-PLUS/Gaia magnitude limits remove the faintest (reddest) dwarfs, so the sample is not volume-complete near 500 pc. The deficit at d < 50 pc reflects both the small volume and the bright-star saturation removed by FLAGS < 4.

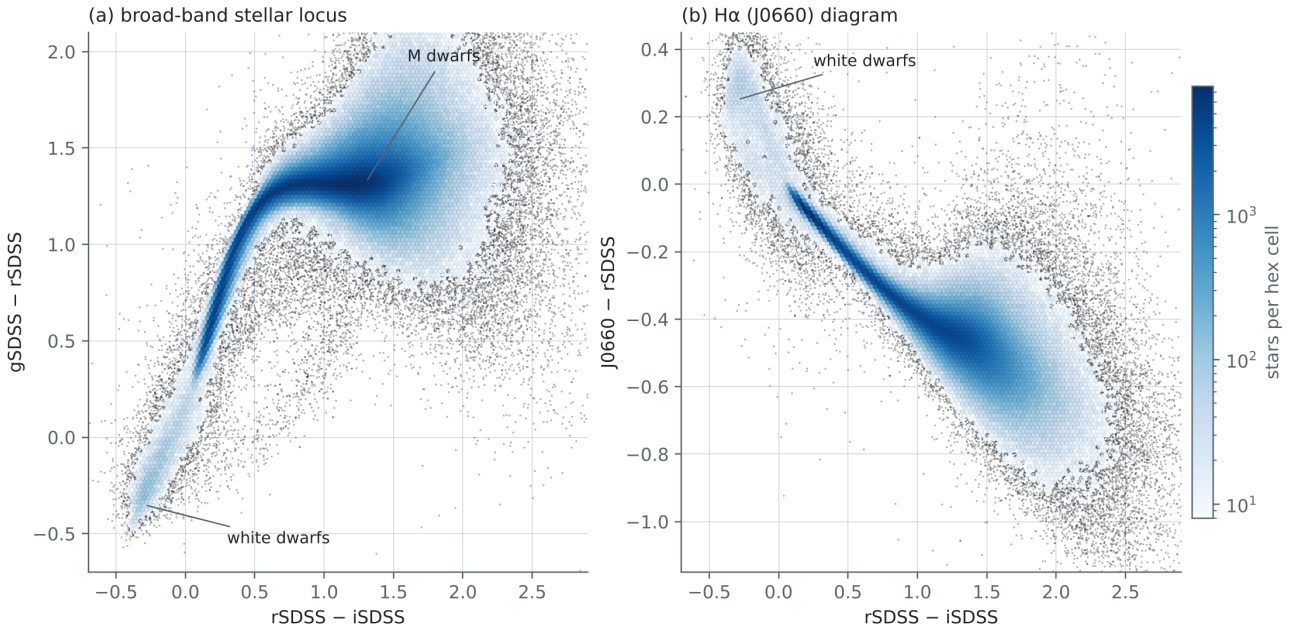


Figure 2. The two colour–colour diagrams. Hexagons show density (logarithmic); grey dots are individual stars in sparse regions. (a) g − r vs r − i: the main-sequence locus bends to a near-constant g − r ≈ 1.3–1.4 for M dwarfs; white dwarfs form the separate sequence at the blue end. The vertical spread at r − i > 1 is largely photometric noise in g, where M dwarfs are faint. (b) J0660 − r vs r − i: J0660 samples Hα. White dwarfs sit at the top left (broad Balmer absorption makes J0660 − r red); chromospherically active M dwarfs with Hα emission are expected below the main locus at r − i > 1.

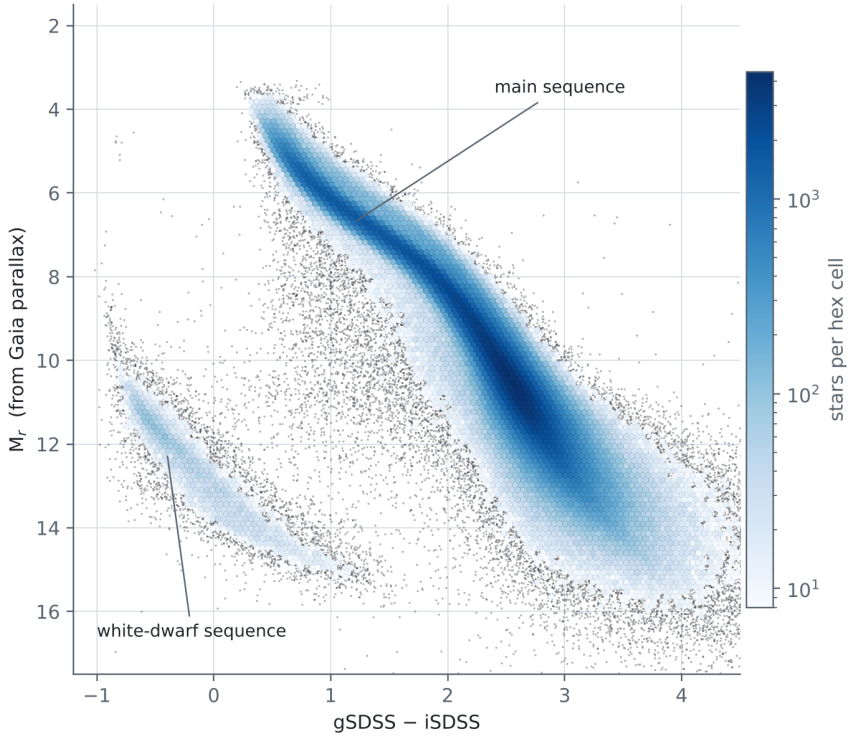


Figure 3. Absolute-magnitude diagram, $M_r = r + 5 \log_{10}(\varpi/100\ \mathrm{mas})$, as a check on the distances. Both the main sequence and the white-dwarf sequence are narrow and well separated, confirming that the parallax selection is sound. The sparse stars between the two sequences are most likely white dwarf + M dwarf binaries or sources with poor astrometry; a RUWE cut would remove part of them.

6 Caveats and possible next steps

- **Extinction** was not corrected; at d < 500 pc it is small at high latitude but not zero near the Galactic plane (J-PLUS provides `jplus.MWExtinction`).

- **Astrometric quality:** 93% of the sample has RUWE < 1.4; adding that cut is a one-line change (the column is in the files).
- **Photometric errors:** no error cut was applied beyond the flags; restricting err_g < 0.05 mag would tighten the M-dwarf part of Figure 2a.
- **Parallax zero point** (≈ −0.017 mas in Gaia DR3) is negligible compared with $\varpi > 2$ mas and was ignored.
- **Other diagrams:** any of the 12 filters can be combined; e.g. J0395 − J0410 (Ca II H+K, metallicity / activity) or u − g (white dwarfs, QSO contamination).

7 Files and reproducibility

File	Content
`jplus_nearby_ccd.py`	Full pipeline: chunked TAP query → cuts → clean catalogue → figure. `python jplus_nearby_ccd.py` (≈ 5 min); `--replot` redraws from the saved raw file.
`jplus_dr4_nearby_clean.fits.gz`	126 MB. 1,089,852 stars, 35 columns: gaia_source_id, ra, dec, tile_id, number, plx, e_plx, dist_pc, ruwe, gaia_g, gaia_bp_rp, and mag_/err_ for all 12 filters in wavelength order. A magnitude is NaN where that filter fails the quality cuts.
`jplus_dr4_nearby_raw.fits`	460 MB. All 1,553,253 Gaia-matched rows before the J-PLUS cuts, with the original 12-element array columns.
`jplus_nearby_ccd.png`	Three-panel summary figure (both diagrams + CMD).

Software: Python 3, pyvo, astropy, numpy, matplotlib. Data: J-PLUS DR4 (CEFCA) and Gaia DR3 (ESA), via the J-PLUS cross-match table. Analysis carried out on 5 November 2026 with the help of Claude (Anthropic).